

Capstone Foundation — Assignment 1

Domain Data & Knowledge Microcosm Blueprint

Your team will build one small but realistic Domain data and knowledge microcosm and reuse it throughout the semester. Every later assignment will extend this same microcosm—from structured storage and governance to knowledge retrieval, explainability, evaluation, and business decision support.

Course Domain Options	Focus
FinTech — Credit & Loan Review	Loan decisioning, underwriting rules, fair-lending/compliance knowledge, and credit-event data.
HealthTech — Medical Claims & Payer Auditing	Claims review, plan coverage rules, procedure-code relationships, payer policies, and auditing knowledge.

The goal of the Microcosm is not to create the largest or most technically sophisticated dataset. The goal is to create a coherent microcosm that is rich enough to test architecture patterns and business consequences across the semester.

This assignment focuses on designing and assembling the domain microcosm. **Teams are NOT** expected to implement the downstream data pipeline, database or any functionality at this point. You don't need huge files, approximately 100-1,000 rows per file is sufficient.

1. Team Structure and Ownership

Teams and domains will be assigned by the instructor after the first class. Roles indicate primary ownership—not exclusive responsibility. Every student is expected to understand the complete end-to-end architecture and participate in major design decisions. Assignment 4 later in the semester will be completed individually. Each team should select a Team lead/liaison (note: the liaison is a secondary responsibility).

Primary Role	Primary Ownership
Data & Ingestion Lead	Source files, ingestion flow, rerun/idempotency strategy, and movement of data through the pipeline.
Database & Analytics Lead	Relational model, DuckDB/SQL work, keys and relationships, curated structured outputs, and point-in-time logic.
Knowledge & Retrieval Lead	Document corpus, chunk/metadata design, embeddings, vector storage, retrieval rules, and source traceability.
Quality & Governance Lead	Validation, data contracts, quality gates, lineage, freshness/authority, access metadata, and explainability controls.

2. Purpose of Assignment 1

Create small but realistic business data and knowledge microcosm that represents the operating environment for your assigned domain. The microcosm will be the common foundation for all subsequent assignments and the final capstone.

Across the semester, the work will progress through the following architecture:

**Structured Data → Ingestion → Storage → Validation & Governance → Knowledge Representation → Metadata
→ Retrieval → Explainability → Business Decision Support**

3. Part 1 — Define the Business Scenario

Before creating files, define the business context that the microcosm represents. Your blueprint must identify:

- The business function or process being represented.
- The primary users or stakeholders of the future solution.
- The business problem the data/knowledge environment should help address.
- One primary decision question the final solution should help answer.
- Two or three supporting questions that may be explored during the semester.

4. Part 2 — Structured Data Package

Create at least three related CSV files representing the structured operational or transactional layer of your domain.

Requirement	Minimum Standard
Number of files	At least 3 related CSV files.
Data roles	At least 2 entity/reference datasets and at least 1 transaction/event dataset.

Requirement	Minimum Standard
Relationships	Primary and foreign-key relationships must be identifiable.
Time element	At least one dataset must include a date/time field suitable for historical or point-in-time analysis.
Scale	Approximately 100–1,000 rows per file is sufficient. Large datasets are not required.
Creation method	Excel/Sheets, synthetic generation, public data adaptation, or a combination. Coding is optional for Assignment 1.

Required structured-data documentation

- Entity Relationship Diagram (ERD): show tables/files, primary keys, foreign keys, major relationships, and what one row represents in each dataset.
- Structured Data Dictionary: for key fields, identify dataset/table, field name, data type, business definition, key status (PK/FK), and whether the field is required or optional.

5. Part 3 — Unstructured Knowledge Corpus

Create or source the document corpus that will later support knowledge representation, chunking, metadata design, vector retrieval, freshness/authority rules, and evidence-based outputs.

Corpus Requirement	Minimum Standard
Number of documents	6–10 distinct documents.
Format diversity	At least 3 multi-page, text-based PDFs and at least 3 TXT or Markdown files.
Structural challenge	At least 1 document with a substantial table and at least 1 document with hierarchical headings or lists.
Version/state diversity	At least 2 similar or related documents representing different states (e.g., CURRENT vs. SUPERSEDED, APPROVED vs. DRAFT, 2026 vs. 2024).
Content suitability	Documents must contain knowledge that can plausibly support the business questions defined in Part 1.

Permitted sources: public documents, government or regulator websites, public policies/manuals, synthetic internal memos, or other instructor-approved sources. If an LLM is used to generate synthetic internal documents, label them clearly as synthetic.

6. Part 4 — Document Metadata Inventory

Create a metadata record for every document in the corpus. Metadata will become the control layer used later for governance, filtering, freshness, authority, lineage, and explainable retrieval. Every document in the corpus must include the metadata fields listed below. These are the minimum required fields for all teams. Teams may add additional domain-specific metadata fields when useful.

Required Field	Purpose / Example
doc_id	Unique document identifier (e.g., FIN-POL-001).
title	Human-readable document title.
doc_type	Policy, procedure, guideline, FAQ, memo, contract, benefits booklet, etc.
source	Public URL/source organization or “Synthetic Internal.”
owner	Business unit or function responsible for the document.
effective_date	Date the document becomes applicable.
version	Version number or label.
jurisdiction / region	Where the document applies, if relevant.
authority_status	Current, approved, draft, superseded, expired.
confidentiality	Public, internal, confidential.
access_group	Who should be permitted to retrieve/use the document.

7. Part 5 — Connect Structured Data to Business Knowledge

Your microcosm must make it possible to connect structured business events to relevant knowledge. Identify at least two examples showing how a structured field, event, or classification should lead a user or AI system to the correct policy, rule, guideline, or procedure.

Example:

Structured Signal	Knowledge Needed	Why the Link Matters
Procedure code + plan code	Current plan coverage/exclusion language	Determines whether a claim decision is consistent with plan rules.

8. Part 6 — Controlled Data and Knowledge Quality Problems

Intentionally include at least three controlled quality problems in the microcosm. These will be used later for validation, quarantine, governance, and retrieval exercises.

Example:

Category	Include at Least One Example
Document/format issue	Missing metadata, poorly formatted table/list, incomplete text, or document that requires review before indexing.

9. Part 7 — Business Risk Matrix

Create a table mapping at least three data/knowledge problems to their downstream AI and business consequences. The purpose is to connect technical architecture decisions to business risk.

Example:

Data / Knowledge Issue	Architecture Layer	Potential AI Impact	Business Impact	Future Control
Duplicate claim/loan event	Ingestion	Incorrect counts or duplicate evidence	Incorrect decision/audit result	Idempotency and deduplication

10. Domain Briefs

Domain 1 — FinTech: Credit & Loan Review

Business problem: streamline credit/loan review while applying current underwriting criteria, fair-lending requirements, product rules, and internal risk limits.

Component	Suggested Foundation
Structured data	applicants.csv — applicant_id, income, employment duration, credit/risk tier, region, requested amount.
Structured data	loan_products.csv — product_code, eligibility criteria, minimum credit requirements, base rate, maximum amount.
Structured data	credit_pull_events.csv — event_id, applicant_id, event_date, credit score/tier change, source/status.
Knowledge corpus	Public fair-lending guidance; consumer lending guidance; internal/synthetic underwriting policies; product eligibility rules; regional/state lending requirements; approval/escalation procedures.
Required linkage	Show how an applicant profile/risk tier and requested loan/product connect to the current underwriting and compliance documents that determine applicable limits or rules.

Domain 2 — HealthTech: Medical Claims & Payer Auditing

Business problem: review claim/billing events against applicable plan and policy knowledge to identify inconsistencies, errors, or potentially incorrect claim decisions.

Component	Suggested Foundation
Structured data	patient_accounts.csv — synthetic member/policyholder ID, enrollment dates, plan code, region, eligibility status.
Structured data	procedure_catalog.csv — standardized or simplified procedure/diagnosis codes, category, description, review flags.
Structured data	claims_ledger.csv — claim_id, member_id, procedure code, submission date, billed/allowed amount, decision/status.
Knowledge corpus	Public coverage guidance; synthetic plan documents; benefits booklets; payer policies; employer-plan addenda; state or federal healthcare guidance relevant to the scenario.
Required linkage	Show how a claim event, plan code, and procedure code connect to the current policy documents that contain coverage, exclusion, or review requirements.

11. GitHub Repository Requirements

Each team will use the following standardized GitHub repository structure. For Assignment 1, populate the following minimum folders/files:

```

team-domain-project/
├── README.md
├── data/
│   ├── structured/
│   │   └── raw/                # 3+ CSV files
│   ├── documents/
│   │   └── source/            # 6-10 source documents
├── metadata/
│   ├── structured_data_dictionary.csv
│   └── document_metadata.csv
├── architecture/
│   └── domain_microcosm.png
└── deliverables/
    ├── assignment_1/
    └── Domain_Microcosm_Blueprint.pdf (or .md/.docx)

```

README.md must identify: team members and primary roles, assigned domain, business scenario, primary decision question, and a short description of the microcosm.

12. Written Blueprint Deliverable

Submit a concise 2–3 page narrative blueprint, excluding diagrams, inventories, data dictionaries, and risk-matrix appendices. The written blueprint must contain these five sections:

1. Business Scenario & Decision Question — context, stakeholders, primary decision question, and supporting questions.
2. Structured Data Architecture — datasets, keys/relationships, ERD reference, and the intended analytical use of the data.
3. Knowledge Corpus & Metadata — document inventory, versions/states, document types, and metadata.
4. End-to-End Domain Microcosm Diagram — a simple diagram showing the structured and knowledge assets and how they relate to the business problem.
5. Business Risk Matrix — at least three risks showing how data/knowledge problems can affect downstream AI outputs and business decisions.

13. Submission Checklist

- ☐ Business scenario, primary decision question, and supporting questions are defined.
- ☐ At least 3 related structured CSV files are included.
- ☐ At least 2 entity/reference datasets and 1 transaction/event dataset are present.
- ☐ ERD and structured data dictionary are included.
- ☐ 6–10 source documents are included: at least 3 PDFs and 3 TXT/Markdown files.
- ☐ Corpus includes at least one substantial table and one hierarchical document structure.
- ☐ Corpus includes at least two related documents with different versions/states.
- ☐ document_metadata.csv contains the required governance/retrieval fields.
- ☐ At least two structured-data-to-knowledge linkages are identified.
- ☐ At least three controlled quality/governance problems are intentionally included.
- ☐ Business Risk Matrix includes at least three risks and future controls.
- ☐ GitHub repository follows the provided structure.
- ☐ No real PHI, PII, confidential customer data, or confidential employer/client data is used.

